

RIZWAN AHAMED H

Software Engineering Intern | Computer Science Undergraduate

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SUMMARY

Software Engineering Intern candidate and Computer Science undergraduate currently pursuing a Bachelor of Engineering (CGPA 8.4/10, expected 05/2028) with one year of object-oriented programming experience in Java, Python, C++, and JavaScript. Strong foundation in data structures, algorithms, software design, testing, distributed systems, and product quality. Built production software used by real users, translating stakeholder and user requirements into reliable, observable, efficient, and scalable technical solutions.

EDUCATION

Dr. Mahalingam College of Engineering and Technology, Pollachi 08/2024 - 05/2028
B.E. Computer Science and Engineering | CGPA: 8.4/10.0 | Currently pursuing Bachelor's degree
- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Probability and Statistics
- Programming practice: Solved 100+ LeetCode problems across arrays, strings, trees, graphs, and dynamic programming
Akshaya Academy CBSE School - TAAC, Palani Graduated 05/2024
- HSC (Class XII): 83% | SSLC (Class X): 91%

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript, SQL
CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming, Software Design, Problem Solving, Concurrency
Backend and APIs: Spring Boot, FastAPI, REST APIs, Distributed Systems, Networking, RBAC, JWT
Databases/DevOps: MySQL, PostgreSQL, SQLite, MongoDB, Docker, Jenkins, Nginx, Linux, Git
Quality/Ops: Testing, Monitoring, Observability, Reliability, Availability, Performance, Security, Code Review
AI/ML: Machine Learning, Ollama, LangChain, Sentence Transformers, Prompt Engineering

PROJECTS

Quad - Self-Hosted Campus Developer Platform 01/2025 - Present
Python, FastAPI, Docker, SQLite, Nginx, Ollama
- Architected a distributed systems stack handling 150+ concurrent projects on 16GB RAM; auto-detects stacks from project manifests, generates multi-stage Dockerfiles, and schedules containers with scale-to-zero for availability, reliability, efficiency, and performance.
- Implemented reverse-tunnel networking, per-project HTTPS, environment secrets, rollback, build log capture, rate limiting, synchronization, and path-traversal hardening; maintained 220+ unit, integration, and security tests.
- Learned and adopted local LLM methods with Ollama to support on-premise codebase Q and A, faculty submission analysis, and code health scoring while reducing SaaS dependency risk.
SACL PMS - Production Quality Control System 08/2024 - 01/2025
Java, Spring Boot, MySQL, Docker, RBAC
- Designed, developed, tested, and deployed a live production software system for Sakthi Auto Components Ltd. automotive OEM servers, replacing paper QC registers across 4 inspection workflows.
- Worked from operator, quality approver, and admin user requirements to build 3-role RBAC, JWT authentication, approval state machine, audit trails, and reliable data capture for product quality operations.
SafeDeploy - Pre-Deployment Simulation Platform 01/2025 - 02/2025
Java, Docker, Jenkins, Linux
- Built a pre-deployment simulation tool for CPU, memory, and I/O constraints to improve software testing, debugging, and reliability before production release.
- Used monitoring and debugging to identify a Java heap memory leak that could have caused production downtime, then applied feedback to improve technical solution quality.

EXPERIENCE

Executive Member - Technical Staff | Student Research Club (SRC), Dr. MCET 01/2025 - Present
- Sole developer of src.drmcet.ac.in; build the live website, internal tools, and lab server operations for a 500+ member technical community.
- Collaborate with students, faculty, and technical teammates to prioritize features, review requirements, and ship reliable internal software.
Vice President and Technical Lead | Ignite Association, Dr. MCET 08/2024 - 01/2025
- Led technical execution for a 1,200+ member association, coordinating event infrastructure, registrations, and cross-department stakeholder needs.
- Drove 40%+ growth in participation by improving scalable event software, communication workflows, and accountability across the team.

ADDITIONAL STRENGTHS

Growth mindset, collaboration, accountability, creative engineering, time management, technical guidance, feedback-driven iteration, and cooperative team delivery.